International Islamic University Chittagong

Center for General Education (CGED)

Midterm Examination, Spring–2025

Course Code: URED-2302 Course Title: Sciences of Qur'an and Hadith

(For Law faculty: URED-2101)

Full Marks: 30 ***Time: 1 hour &30 minutes***

Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question.

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	Define Al-Qur'an literally and terminologically. Explain some extraordinary characteristics of the holy Qur'an with proper evidence.	10	2	Remember & Create
2	Define Wahi literally and terminologically. Explain some types of Wahi elaborately. Prove the Wahi as the best source of Shari`ah.	10	2	Remember & Create
3	(a) "The arrangement of Ayah and Surah or the holy Qur'an is by the command of Prophet (SAAS)"- justify this statement summarizing the opinion of Muslim scholars. **Or,** (b) Define Ayah and Surah literally and terminologically. Explain the classifications of the Surah of the holy Qur'an elaborately.	10	2	Evaluate & Create

Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

Mid Term Examination, Spring 2025

CSE 2321 Data Structures

Total marks: ***30*** Time: ***90*** minutes

[Answer *all* of the following questions. Figures in the right-hand margin indicate full marks.]

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1. a) What do you mean by a *data structure*? Briefly describe the different operations performed on data structures. 2 CO1, C2

b) Draw a *flowchart* for *linear search* algorithm. 3 CO1 C2

[Or]

Draw a *flowchart* to find the *smallest element* of an array.

c) What is the *time complexity* of the following functions in Big O notation? Explain. 2 CO3 C4

i)

```
int solve1(int n){
    int i = n;
    int cnt = 0;
    while(i>0){
        i = i /2;
        cnt++;
    }
    return cnt;
}
```

ii)

```
int solve2(int n){
    int cnt = 0;
    for(int i=0;i<n;i++){
        for(int j=1; j<n; j*=2){
            cnt = cnt + 1;
        }
    }
    return cnt;
}
```

d) Given the pattern **P = 'ccd'** and text **T = 'cbcddcddcdcccddd'**, determine the number N of character comparisons required to find the first occurrence (index) of P in T using the naive *pattern matching algorithm*. Show the matching process step by step until the pattern is found. 3 C03 C3

2. a) Write an algorithm to *insert* an element **ITEM** to the **K**th position of a linear array *LA*. What is the time complexity of this algorithm? 2 CO1 C2

b) Consider a 2D array **A(4:12, 7:14).** 2.5 C03 C3

i. Find the length of each dimension and the total number of elements in A.

ii. If **Base(A) = 500** and **w = 4** bytes per memory cell, calculate the address of the element **A[6, 10]** assuming row-major storage order.

c) Given a string **BRIDGE,** find the number of *comparisons* (C) and number of *interchanges* (D) needed to *sort* the string *alphabetically* by using *bubble sort* algorithm. Show each steps. 4 CO3 C3

[OR]

You are developing a search feature for a library management system that stores book IDs in a sorted list. To enable quick look-ups, which search algorithm would you choose to efficiently check if a given book ID exists? Write the algorithm for this search.

d) Write down the *worst case* & *average case* time complexity for the following algorithm: 1.5 C01 C1

i) Binary search ii) Linear search iii) Bubble Sort

3. a) What is *linked list*? What is the difference between an array and a linked list in terms of their memory structure and operations? Which one is more efficient for searching an element, and why? 2 CO1 C1

b) Describe how you can *insert* a new node at the middle position of a singly linked list. Write an **algorithm or C/C++ function** for the insertion, and state the time complexity of this insertion operation. 4 CO3 C3

[OR]

You are developing a **music playlist manager** using a **singly linked list.** Each node in the list stores the name of a song. The playlist contains an **odd number of songs,** but the number of songs is **unknown.** Write an **efficient algorithm or C/C++ function** to find the **middle song** of the playlist using **only one traversal.**

c) Given is the following sorted linked list where the array **INFO** contains a list of integer numbers and **LINK, START** and **AVAIL** are the pointer fields- 2 CO1 C2

	1	2	3	4	5	6	7	8	9	10	11	12
INFO:			41	129		9	123	78	194	231	62	145
LINK:	2	5	11	12	0	3	4	7	10	0	8	9

START : 6 AVAIL : 1

Determine the changes in the list if number **XX** is *added* to the list and then **145** is *deleted* from the list.
[Here XX is the last two digits of your ID]

d) What is *two way linked list*? Form a *two way linked list* from the one way list in **Q 3(c).** 2 CO1 C3

International Islamic University Chittagong

Department of Computer Science & Engineering

Program: B.Sc. in CSE: Semester: 3rd

Mid Term Examination, Spring-2025

Course Code: CSE-2323 | Time: 1 Hour 30 minutes.

Course Title: Digital Logic Design | Total Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

Question No	Question	CLOs	Marks
1.a	What do you mean by Parity code? If you receive a hamming code '1101110' with even parity then detect and correct the error.	**CLO1**	5
1.b	Write down the half adder's truth table and its output Equation. and use AOI gates to implement it. Afterward, change AOI to NAND gates.	**CLO2**	5
2.a	Simplify the Boolean function $F(A,B,C,D,E)=\sum(0,2,3,4,5,6,7,11,15,16,18,19,23,27,31)$	**CLO2**	4
2.b	Convert the Boolean expression in canonical form i. $F= AB'+A'BC+ A'B'C'D$ **(SOP form)** ii. $(A' + B).(B' + C + D')(A'+B'+C+D)$ **(POS form)**	**CLO1**	4
2.c	Add 8867 with 2832 using BCD addition	**CLO1**	2
3.a	**Simplify** and **Implement** the function F .which has don't care condition d. $F(A,B,C,D)=\sum(0,1,2,3,7,8,10)$ $d(A,B,C,D) = \sum(5,6,11,15)$.	**CLO2**	4
	OR		
3.a.	In a room, there are three switches (S1, S2, and S3) controlling a light. The light will turn on (output = 1) under the following conditions: 1. The light turns on if **S1** is on (1) and **S2** is off (0), regardless of S3. 2. The light turns on if **S3** is on (1) and **S2** is on (1), regardless of **S1**. 3. The light turns off (output = 0) in all other cases.	**CLO2**	4

	Task: Design a combinational logic circuit to control the light, based on the above conditions. Derive the truth table and Simplify it using Karnaugh Map (K-map).		
3.b	An oxygen control system has 4 inputs: • A = Oxygen tank level is low (low=1,otherwise=0) • B = Backup tank available(available=1,not available=0) • C = Patient is critical (critical=1,otherwise-0) • D = Manual override switch (switch on=1,off=0) Oxygen must be **supplied (Y = 1)** if: • Patient is critical, regardless of anything • OR manual override is on • OR tank is low **AND** backup is available **Tasks** i.Derive the truth Table ii.Derive the logical expression of Y iii.Draw the logic circuit	**CLO2**	6

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
B. Sc. in CSE, Mid Term Examination, Spring-2025
Course Code: MATH-2307, Course Title: Mathematics-III

Time: 1:30 hours **Marks: 30**

[Answer all questions; Read the instruction below before answering the questions; Figures in right hand margin indicate full marks]

1\. a) If the matrix $A = \begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is symmetric, find the value of x and hence present the matrix A — 3 CLO1 C2

Or

Prove that $A=\begin{bmatrix} 2 & 2-3i & 3+5i \\ 2+3i & 3 & i \\ 3-5i & -i & 5 \end{bmatrix}$ is Hermitian — 3 CLO1 C2

b) Check whether it is singular matrix $\begin{bmatrix} 1 & 2 & d \\ 2 & 4 & e \\ 3 & 5 & f \end{bmatrix}$; Where d , e and f are the digits of your own ID number — 2 CLO1 C2

c) Name 5 methods of finding the inverse of a matrix — 2 CLO1 C2

d) Find the dependent row(s) from the following linear system of equations — 1 CLO1 C2

$2x+3y+z=1$

$4x+5y+7z=9$

$x+9y+z=0$

$4x+6y+2z=2$

e) Find the area of the parallelogram constructed by the vectors $\vec{u} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ — 2 CLO1 C2

Or

Test whether $\lambda_1 = f$ *and* $\lambda_2 = e$ are Eigen values for $A = \begin{bmatrix} 5 & 3 \\ 2 & 3 \end{bmatrix}$; Where e and f are the digits of your own ID number — 2 CLO2 C2

2\. a) Determine which of the following vectors are Eigen vectors for $A = \begin{pmatrix} 2 & f \\ -1 & -5 \end{pmatrix}$ showing your analysis procedure graphically — 5 CLO2 C2

(a)$\begin{pmatrix} a \\ e \end{pmatrix}$ (b)$\begin{pmatrix} b \\ f \end{pmatrix}$; Where a , b, e and f are the digits of your own ID number

b) **Determine whether the following vectors $\{(2, 1, 2), (0, 1, -1), (4, 3, 3)\}$ in R^3 are linearly dependent or linearly independent** 5 CLO2 C2

Or

Verify Cayley Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$ **and hence find** A^{-1}**.** 5 CLO2 C2

3. **Diagonalize the matrix** $\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$ 10 CLO2 C2

Instruction:

If your ID no is C171023 then you can assign your digit as follows

C	1	7	1	0	2	3
	↓	↓	↓	↓	↓	↓
	a	b	c	d	e	f

To solve the above problems, you need to use your own ID number where needed

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE, Mid Terml Examination, Spring-2025

Course Code:CHEM- 2301 Course Title: Chemistry

Total marks: ***30*** Time: ***1*** hours ***30*** minutes

[Answer ***all*** the questions. Figures in the right hand margin indicate full marks.]

Bloom`s Levels of the Questions

Letter Symbols	R	Un	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

			BT	Marks
1.	a)	Write down the modern periodic law and mention some properties of modern periodic table. Discuss the classification of elements based on electronic configurations with examples.	R/Un	2+3
1.	b)	"Electronic Configuration of the elements is the basis of the Periodic Table"- explain.	An	2
1.	c)	What is electronegativity? Why Fluorine (F) has the highest electronegativity?	R/Un	1+2
		OR		
1.	a)	What is quantum number? Mention four quantum numbers with their significance.	R/Un	1+2
1.	b)	Determine position of the following elements in the periodic table. (i)**Cu_{29}** **(ii) Br_{35}**	Ap	2
1.	c)	Compare Rutherford's Model with Bohr's Model. Which model is more acceptable and why?Mention with their figures.	An	2+3
2.	a)	What do you mean by Coordination covalent bond? Explain with an example.	Un	2
2.	b)	Analyze and show clearly- the bond formation in NaCl.	An	3
2.	c)	Explain Hybridisation process in **$BeCl_2$ & H_2O** .Mention bond angle ,structure.	Ap/ An	3+2
3.	a)	What is transition Elements? Explain why all d-block elements aren't transition elements but all transition elements are d-block elements?	R/Un	1+2
3.	b)	What do you mean by Lanthanides & Actinides contraction?	Un	2
3.	c)	What are the properties of transition elements?Explain briefly three of them.	R/An	2+3

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Spring- 2025

Course Code: STAT 2311 Course Title: Probability and Statistics

Total marks: 30, Time: 1 hours 30 minutes

[Answer all the questions; Figures in the right hand margin indicate full marks.] CO DL

1.

a) What is a questionnaire? 2 CO1 C1

b) Design a structured questionnaire with at least 20 questions on the topic: "Factors Affecting Adverse Reactions of COVID-19 Vaccine in Bangladesh." 8 CO1 C6

2.

a) What is the geometric mean? How is it different from the harmonic mean? Explain the applications of various measures of central tendency. 5 CO1 C2

b) The boot time (in seconds) of 20 different systems in a lab is recorded as follows: 5 CO1 C4

Boot Time (sec)	5–10	10–15	15–20	20–25	25–30
No. of Systems	2	5	7	4	2

(i) Calculate the first quartile (Q_1), ninth decile (D_9), and eighty-seventh percentile (P_{87}) from the given data.

(iv) Draw an appropriate cumulative frequency curve (ogive) and use it to locate the values of Q_2, D_9, and P_{71}.

3.

a) What are the various types of measures of dispersion in statistics? Explain any two with appropriate examples. 5 CO1 C2

b) The accuracy rates (in percentage) of two machine learning models, Model A and Model B are as follows: 5 CO1 C4

Model A	89	92	85	91	88	90
Model B	86	88	87	84	89	**Your weight (in kilograms)**

(i) Which model has a higher average accuracy?

(ii) Which model demonstrates greater consistency in its accuracy rates?

Or

a) Define a variable with an example, and explain how a discrete variable differs from a continuous variable. 5 CO1 C2

b) Write your birthdate in ddmmyyyy format (with a zero before single-digit day or month if needed), separating each digit with commas. Then, calculate the coefficient of variation for these digits. 5 CO1 C4